

IC commands & messages 정리

ICS commands (CCD 제어 상위 기능/명령어)

STATUS: ICS 설정 상태 반환

ACQSTATUS: IC K/M/T/N과 연결 및 각 IC 초기화 상태 반환

FILENAME: 현재 설정되어 있는 ICS 자체의 파일이름 반환

SYNCHRONIZE ((return IMGTYPE/OBJNAME/EXP/OBSERVER/PROJID)): IMGTYPE/OBJNAME/EXP/OBSERVER/PROJID의 현재 설정 값 반환, 주로 IC K/M/T/N 실행되면 초기화를 위해 각 IC에서 ICS로 명령어를 보냄.

EXPNUM <n>: FILENAME에 들어가는 일련번호 설정, 영상파일 이름은 KMTNx.yyyymmdd.nnnnnn.fits로 설정됨 (e.g. KMTNk.20151002.012345.fits, 6자리)

TIME: 현재 OS/FITS/RTC 시각정보 등 반환

BIN <n> ((not implemented)): CCD binning 설정, 명령어 리스트에는 있으나, 명령어 구현되어 있지 않음

ROI ((reserved)): Region of interest (sub section image) 설정, 명령어는 구현되어 있으나, 정상적으로 동작하지 않음

DISPL ((reserved)): 명령어는 구현되어 있으나, 정상적으로 동작하지 않음

LEDFLASH <x>: 장비 점검용 LED 점등시간 설정, 0으로 설정하면 LED 켜지지 않음

PROJID <project ID>: 관측 프로젝트 ID 설정, FITS 헤더에 들어가는 정보로 CCD 구동 설정에는 영향 없음

OBSERVER <observer's name>: 관측자 이름 설정(띄어쓰기 허용), OBSERVER 이후 문자열이 헤더에 들어감, CCD 구동 설정에는 영향 없음

EXP <x.x>: CCD 노출시간(EXPTIME, exposure time) 설정

BIAS <object name>: Image type(FITS header에 IMAGETYP 키워드로 들어감)을 BIAS로 설정하고, Object name(FITS header에 OBJECT 키워드로 들어감)을 <object name>에 입력한 문자열로 변경(e.g. "BIAS zero" 입력 → IMAGETYP="BIAS" / OBJECT="zero"), <object name> 없이 BIAS만 입력하면 Image type만 변경 됨. Image type이 BIAS로 설정되면, 노출시간은 0초이고 셔터도 열리지 않음.

DARK <object name>: Image type을 DARK로 설정하고, Object name 변경, <object name> 입력 안하면 Image type만 변경, DARK로 설정되면 셔터가 열리지 않음.

OBJECT <object name>: Image type을 OBJECT로 설정하고, Object name 변경, <object name> 입력 안하면 Image type만 변경, 설정된 노출시간에 따라 셔터 열림.

FLAT <object name>: Image type을 FLAT으로 설정, 나머지는 OBJECT 커맨드와 동일

SKY <object name>: Image type을 SKY로 설정, 나머지는 OBJECT 커맨드와 동일

DOMEFLAT <object name>: Image type을 DOMEFLAT으로 설정, 나머지는 OBJECT 커맨드와 동일

STANDARD <object name>: Image type을 STANDARD로 설정, 나머지는 OBJECT 커맨드와 동일

GO : 노출 1장 시작, configuration - CCD flushing - integration - shutter wait - readout - FITS building/writing 이 순차적으로 진행되며, 현재 진행상태를 계속 화면에 출력 및 명령을 내린 호스트에게 보고

GO <n> : n장 노출 반복, Go 명령을 n번 내릴 것과 동일한 기능

STOP ((not implemented)): 노출(integration) 중지 후 리드아웃 및 영상저장, 현재 명령어 리스트에는 있으나, 명령어 구현되어 있지 않음

ABORT ((not implemented)): 노출 및 관련 프로세스 중지, 리드아웃 및 영상저장 안함, 현재 명령어 리스트에는 있으나, 명령어 구현되어 있지 않음

MOVIE ((not implemented)): 현재 명령어 리스트에는 있으나, 명령어 구현되어 있지 않음

* ICS>TC AUXSTATUS: TCSAgent에 AUX status 요청

* ICS>TC TCSSTATUS: TCSAgent에 TCS status 요청

※ 각 설정 명령어에서 매개변수를 입력하지 않으면 현재 설정되어 있는 값을 반환

※ 몇몇을 제외한 대부분의 설정 명령어는, ICS에서 다시 각 IC에 그대로 명령을 전달하여 모든 IC의 설정을 동기화 하는 기능

IC K/M/T/N commands

STATUS: IC K/M/T/N 설정 상태 반환

DMAWAIT <n>: 자료 저장 오류를 줄이기 위한, Optical fiber 통신 지연 설정

DATASOURCE <ADC/CTC>: onboard crosstalk 보정 옵션 설정 (ADC: ADC 출력 그대로 전송, CTC: corss-talk correction 하여 전송)

LEDFLASH <x> ((no function)): LED flash 시간 단위 설정, 현재 기능하지 않음

FILENAME: IC K/M/T/N에서 저장될 파일 이름 반환

* K/M/T/N.IC>ICS SYNCHRONIZE: ICS로 'SYNCHRONIZE' 명령을 보내 회신되는 설정 정보로 IC K/M/T/N 자신을 동기화 설정 (회신메세지 예: "ICS> K/M/T/N.IC DONE: SYNCHRONIZE IMGTYPE=DARK OBJNAME=end EXP=30 OBSERVER=choi_jeong_sik PROJID=OBS"), ">ICS SYNCHRONIZE" 명령이 선행되지 않아도, ICS 뿐만 아니라 제3의 호스트에서 "DONE: SYNCHRONIZE ..." 메시지가 들어오면 동기화 설정이 이루어짐

EXPNUM <n>: FILENAME에 들어가는 일련번호 설정, 각 IC에서 영상파일 이름은 KMTNx.yyyymmdd.nnnn.fits로 설정됨 (e.g. KMTNk.20151002.0123.fits, 4자리), 6자리 일련번호로 설정하려면 INITIALIZE 명령어를 사용해야 함, 보통 ICS에서 날짜와 6자리 일련번호(새로운 노출을 시작할 때 1씩 증가)를 설정하여 각 IC에게 INITIALIZE 명령을 보내 파일명을 동기화함

TIME: 현재 OS/FITS/RTC 시각정보 등 반환

PROJID <project ID>: 각 IC K/M/T/N 자료에 대한 관측 프로젝트 ID 설정, FITS 헤더에 들어가는 정보로 CCD 구동 설정에는 영향 없음

OBSERVER <observer's name>: 각 IC K/M/T/N 자료에 대한 관측자 이름 설정(띄어쓰기 허용), OBSERVER 이후 문자열이 헤더에 들어감, CCD 구동 설정에는 영향 없음

EXP <x.x>: 각 IC K/M/T/N 노출에 대한 CCD 노출시간(EXPTIME, exposure time) 설정

BIAS <object name>: 각 IC K/M/T/N 노출에 대한 image type과 object name 설정, <object name> 입력 안하면 Image type만 변경, Image type에 따라 셔터 구동 여부가 결정됨 (BIAS와 DARK는 셔터 안 열림, BIAS는 노출시간 0초로 설정됨)

DARK <object name>: 상동

OBJECT <object name>: 상동

FLAT <object name>: 상동

SKY <object name>: 상동

DOMEFLAT <object name>: 상동

STANDARD <object name>: 상동

INITIALIZE <suffix>: KMTNx.<suffix>.fits와 같이 설정되는 파일 이름의 <suffix>를 설정, e.g. 20171111.050723 → KMTNk.20171111.050723.fits

ERASE: CCD flushing 시작, IC K(master)에서만 기능

SHOPEN <x.x> <source ID> USESTATUS: 셔터를 <x.x>만큼 open, “<source ID> USESTATUS”를 입력하면 <source ID>에게 STATUS 메시지 형태로 남은 시간을 주기적으로 보내줌, IC K(master)에서만 기능

SHCLOSE: 셔터 close, 보통 SHOPEN으로 설정된 <x.x> 시간이 경과하면 셔터가 닫히도록 되어있지만, 바로 셔터를 닫아 노출을 중단하고 싶을 때 사용, IC K(master)에서만 기능

FLASHNOW <n>: LED를 <n> 시간만큼 점등

GO <source ID>: 각 K/M/T/N CCD readout start, 우선 IC M/N/T에 명령을 내린 후 마지막에 master이 IC K에 명령을 내림

* ICS>K/M/T/N.IC STATUS: AUXSTATUS .. : ICS에서 AUX status 메시지를 보내면 저장해두었다가, FITS 저장 시 header에 정보 기록

* ICS>K/M/T/N.IC STATUS: TCSSTATUS .. : ICS에서 TCS status 메시지를 보내면 저장해두었다가, FITS 저장 시 header에 정보 기록

<Example> -----

LEDFLASH 20000

PROJID ENG

OBJECT led

EXP 10

EXPNUM or INITIALIZE

EXPNUM n (suffix가 yyyyymmdd.####으로 설정됨. ####에 n이 들어감)

INITIALIZE <suffix> (suffix는 아무거나 넣을 수 있음. 관측에서는 20190624.000001와 같이 넣어줌)

ERASE

SHOPEN 10 (or SHOPEN 10 OBS USESTATUS)

FLASHNOW 20000

GO (or GO OBS)

* SHOPEN 전후에 "STATUS: AUXSTATUS .."나 "STATUS: TCSSTATUS .." 날리면 해당정보 입력됨

commands from K.IC to other hosts

K.IC>ICS SYNCHRONIZE

K.IC>XIS TIME

DONE and STATUS message from K.IC

<DARK>

DONE: DMAWAIT DMAWaitTime=500

DONE: DATASOURCE DataSource=ADC

DONE: LEDFLASH LEDFlashTime=1

DONE: PROJID ProjID=OBS

DARK ImageType=DARK ObjectName='begin' EXP=16

DONE: EXP ExpTime=30 seconds.

DONE: INITIALIZE Initialization Complete.

DONE: Erase Cycle Complete.

STATUS: GO

STATUS: GO PCTREAD=xx ((only to sourceID))

STATUS: GO PCTREAD=100 Acquisition Complete. Disk Transfer Starting. ((only to sourceID))

STATUS: GO Acquisition Complete

STATUS: GO Disk Write Complete ((after PONG response from XIS))

DONE: Wrote LASTFILE=/mnt/ICSDData/KMTNk.20171111.050722.fits RATE=408038 KB/sec ((from K.CB))

<OBJECT>

DONE: DMAWAIT DMAWaitTime=500

DONE: DATASOURCE DataSource=ADC

DONE: LEDFLASH LEDFlashTime=20000

DONE: PROJID ProjID=ENG

DONE: OBJECT ImageType=OBJECT ObjectName='led' EXP=30

```

DONE: EXP ExpTime=10 seconds.

DONE: INITIALIZE Initialization Complete.
DONE: Erase Cycle Complete.

STATUS: SHOPEN Shutter=Open (( only to sourceID ))
STATUS: SHOPEN Integration Remaining=xx sec. (( only to sourceID ))
DONE: FLASHNOW LED Flash Done.
STATUS: FLASHNOW Shutter=Closed Integration Remaining=0 sec. (( only to sourceID ))

STATUS: GO
STATUS: GO PCTREAD=xx (( only to sourceID ))
STATUS: GO PCTREAD=100 Acquisition Complete. Disk Transfer Starting. (( only to sourceID ))
STATUS: GO Acquisition Complete

STATUS: GO Disk Write Complete (( after PONG response from XIS ))

DONE: Wrote LASTFILE=/mnt/ICSDData/KMTNk.20171111.050723.fits RATE=1135512 KB/sec (( form K.CB ))

```

ISIS log sample -- extracted for commands between ICS and K.IC (original ISIS log: isis.20171110.log)

XIS runtime log (re)started at UTC 2017-11-11T00:26:37.799322

```

2017-11-11T00:29:00.668493 [192.168.13.102:6600] K.IC>ICS SYNCHRONIZE
2017-11-11T00:29:00.683545 [192.168.13.102:6600] K.IC>XIS TIME
2017-11-11T00:29:00.683591 XIS>K.IC DONE: TIME DATE=2017-11-11 TIME=00:29:00 ISODATE=2017-11-11T00:29:00.683569 TIMESYS=UTC
2017-11-11T00:29:00.684770 [/dev/ttyS0] ICS>K.IC DONE: SYNCHRONIZE IMGTYPE=DARK OBJNAME=end EXP=30 OBSERVER=PJK PROJID=OBS

```

```

2017-11-11T00:34:36.184980 [192.168.13.109:6650] OBS>k.ic dmawait 500
2017-11-11T00:34:36.272281 [192.168.13.102:6600] K.IC>OBS DONE: DMAWAIT DMAWaitTime=500

```

```

2017-11-11T00:34:57.849723 [192.168.13.109:6650] OBS>k.ic datasource adc
2017-11-11T00:34:57.935169 [192.168.13.102:6600] K.IC>OBS DONE: DATASOURCE DataSource=ADC

```

```

2017-11-11T00:35:21.304984 [192.168.13.109:6650] OBS>k.ic ledflash 1
2017-11-11T00:35:21.378349 [192.168.13.102:6600] K.IC>OBS DONE: LEDFLASH LEDFlashTime=1

```

```

2017-11-11T00:35:26.472770 [192.168.13.109:6650] OBS>ICS ledflash 1
2017-11-11T00:35:26.484637 [/dev/ttyS0] ICS>OBS DONE: LEDFLASH LEDFlashTime=1

```

<< DARK with LEDFLASH 20000 >>

```

2017-11-11T00:35:29.408980 [192.168.13.109:6650] OBS>ICS projid obs
2017-11-11T00:35:29.421664 [/dev/ttyS0] ICS>OBS DONE: PROJID ProjID=OBS
2017-11-11T00:35:29.423633 [/dev/ttyS0] ICS>K.IC projid obs

```

```

2017-11-11T00:35:29.494233 [192.168.13.102:6600] K.IC>ICS DONE: PROJID ProjID=OBS

```

```

2017-11-11T00:35:32.672981 [192.168.13.109:6650] OBS>ICS dark begin
2017-11-11T00:35:32.687739 [/dev/ttyS0] ICS>OBS DONE: DARK ImageType=DARK ObjectName='begin' EXP=30
2017-11-11T00:35:32.689731 [/dev/ttyS0] ICS>K.IC dark begin

```

```

2017-11-11T00:35:32.791202 [192.168.13.102:6600] K.IC>ICS DONE: DARK ImageType=DARK ObjectName='begin' EXP=16

```

```

2017-11-11T00:35:34.713721 [192.168.13.109:6650] OBS>ICS exp 30
2017-11-11T00:35:34.727768 [/dev/ttyS0] ICS>OBS DONE: EXP ExpTime=30 seconds.
2017-11-11T00:35:34.729633 [/dev/ttyS0] ICS>K.IC exp 30

```

```

2017-11-11T00:35:34.799100 [192.168.13.102:6600] K.IC>ICS DONE: EXP ExpTime=30 seconds.

```

```

2017-11-11T00:35:39.072725 [192.168.13.109:6650] OBS>ICS go

```

```

2017-11-11T00:35:40.026766 [/dev/ttyS0] ICS>K.IC INITIALIZE 20171111.050722

```

```

2017-11-11T00:35:40.474103 [192.168.13.102:6600] K.IC>ICS DONE: INITIALIZE Initialization Complete.

```

```

2017-11-11T00:35:40.534769 [/dev/ttyS0] ICS>K.IC ERASE (( K only ))

```

```

2017-11-11T00:35:40.756926 [/dev/ttyS0] ICS>K.IC STATUS: AUXSTATUS ENS7=0.0 ENS6=19.2 ENS5=19.2 ENS4=41.7 ENS3=18.8 ENS2=19.2 ENS1=19.3
ENFAN=OFF ENSTAT=STANDBY CHSTAT=NC MCPOS=0 MCSTAT=STANDBY DSSTAT=NC FAPOSW=-4.222 FAPOSE=-7.187 FAPOSS=-5.631 FALIMW=0 FALIME=0
FALIMS=0 FATILTEW=-350.0 FATILTNS=-10.0 FAFOCUS=-5.680 FASTAT=STANDBY SHUTTER=CLOSED SHUTOP=STANDBY FILTER=V FILNUM=3 FILTOP=STANDBY

```

```

FSSTAT=STANDBY AUXUPDATE=2017-11-11T00:35:39.823 AUXARC=Enabled AUXLINK=Up TELID=KMTS TIMESYS=UTC AUXQDATE=2017-11-11T00:35:39.911
KBUILD=KS2016-01-13:1370 MBUILD=KS2016-01-13:1370 TBUILD=KS2016-01-13:1370 NBUILD=KS2016-01-13:1370 GBUILD= ICSBUILD=KX2016-03-
23:1381 EXPSTATUS=ERASE

2017-11-11T00:35:47.775347 [192.168.13.102:6600] K.IC>ICS DONE: Erase Cycle Complete.

2017-11-11T00:35:47.912663 [/dev/ttyS0] ICS>K.IC STATUS: TCSSTATUS DATE-OBS=2017-11-11T00:35:48 EXECODE=E TCSDRIVE=Disabled TCSLIMIT=No
TELMOVE=Idle AZ=0.0 ALT=90.0 SECZ=1.00 ST=05:20:19 HA=+00:00:00 EQUINOX=2000.000 DEC=-32:13:39.5 RA=05:19:44.75 TCSUDATE=2017-11-
11T00:35:40.057 TCSARC=Enabled TCSLINK=Up TIMESYS=UTC TCSQDATE=2017-11-11T00:35:40.760 EXPSTATUS=INTEGRATING

2017-11-11T00:36:17.909765 [/dev/ttyS0] ICS>K.IC GO OBS (( after GO OBS to M.IC/N.IC/T.IC ))
2017-11-11T00:36:17.964061 [192.168.13.102:6600] K.IC>ICS STATUS: GO

2017-11-11T00:36:20.625195 [192.168.13.102:6600] K.IC>OBS STATUS: GO PCTREAD=6 (( only to sourceID ))
2017-11-11T00:36:23.994067 [192.168.13.102:6600] K.IC>OBS STATUS: GO PCTREAD=17 (( only to sourceID ))
2017-11-11T00:36:27.363157 [192.168.13.102:6600] K.IC>OBS STATUS: GO PCTREAD=28 (( only to sourceID ))
2017-11-11T00:36:30.733060 [192.168.13.102:6600] K.IC>OBS STATUS: GO PCTREAD=39 (( only to sourceID ))
2017-11-11T00:36:34.102021 [192.168.13.102:6600] K.IC>OBS STATUS: GO PCTREAD=50 (( only to sourceID ))
2017-11-11T00:36:37.472062 [192.168.13.102:6600] K.IC>OBS STATUS: GO PCTREAD=61 (( only to sourceID ))
2017-11-11T00:36:40.841046 [192.168.13.102:6600] K.IC>OBS STATUS: GO PCTREAD=72 (( only to sourceID ))
2017-11-11T00:36:44.211040 [192.168.13.102:6600] K.IC>OBS STATUS: GO PCTREAD=83 (( only to sourceID ))
2017-11-11T00:36:47.581042 [192.168.13.102:6600] K.IC>OBS STATUS: GO PCTREAD=94 (( only to sourceID ))

2017-11-11T00:36:49.321014 [192.168.13.102:6600] K.IC>OBS STATUS: GO PCTREAD=100 Acquisition Complete. Disk Transfer Starting. (( only
to sourceID ))

2017-11-11T00:36:49.368106 [192.168.13.102:6600] K.IC>ICS STATUS: GO Acquisition Complete
2017-11-11T00:36:49.786650 [/dev/ttyS0] ICS>OBS DONE: EXPSTATUS=IDLE

2017-11-11T00:37:00.372514 [192.168.13.102:6600] K.IC>K.CB TRANSFER DISK1 1 ICS
2017-11-11T00:37:00.389478 [192.168.13.102:6600] K.IC>XIS PING
2017-11-11T00:37:00.389510 XIS>K.IC PONG
2017-11-11T00:37:00.433973 [192.168.13.102:6600] K.IC>ICS STATUS: GO Disk Write Complete (( after PONG response from XIS ))

2017-11-11T00:37:05.790397 [192.168.13.102:10601] K.CB>ICS DONE: Wrote LASTFILE=/mnt/ICSDATA/KMTNk.20171111.050722.fits RATE=408038
KB/sec (( from K.CB ))
2017-11-11T00:37:05.826645 [/dev/ttyS0] ICS>OBS STATUS: Wrote LASTFILE=/mnt/ICSDATA/KMTNk.20171111.050722.fits RATE=408038 KB/sec

2017-11-11T00:37:06.493460 [192.168.13.102:10601] K.CB>K.IC DONE DISK1 1
2017-11-11T00:37:06.493514 [192.168.13.102:10601] K.CB>K.IC REQ SWAP

2017-11-11T00:37:06.580002 [192.168.13.102:6600] K.IC>K.CB ACK SWAP

<< OBJECT with LEDFLASH 20000 >>

2017-11-11T00:38:00.097846 [192.168.13.109:6650] OBS>ICS projid ENG
2017-11-11T00:38:00.110646 [/dev/ttyS0] ICS>OBS DONE: PROJID ProjID=ENG
2017-11-11T00:38:00.112632 [/dev/ttyS0] ICS>K.IC projid ENG

2017-11-11T00:38:00.184183 [192.168.13.102:6600] K.IC>ICS DONE: PROJID ProjID=ENG

2017-11-11T00:38:10.753855 [192.168.13.109:6650] OBS>ICS object led
2017-11-11T00:38:10.768641 [/dev/ttyS0] ICS>OBS DONE: OBJECT ImageType=OBJECT ObjectName='led' EXP=30
2017-11-11T00:38:10.770626 [/dev/ttyS0] ICS>K.IC object led

2017-11-11T00:38:10.869209 [192.168.13.102:6600] K.IC>ICS DONE: OBJECT ImageType=OBJECT ObjectName='led' EXP=30

2017-11-11T00:38:27.073717 [192.168.13.109:6650] OBS>ICS ledflash 20000
2017-11-11T00:38:27.087645 [/dev/ttyS0] ICS>OBS DONE: LEDFLASH LEDFlashTime=20000

2017-11-11T00:38:41.706847 [192.168.13.109:6650] OBS>ICS exp 10
2017-11-11T00:38:41.719645 [/dev/ttyS0] ICS>OBS DONE: EXP ExpTime=10 seconds.

2017-11-11T00:38:41.721639 [/dev/ttyS0] ICS>K.IC exp 10

2017-11-11T00:38:41.786938 [192.168.13.102:6600] K.IC>ICS DONE: EXP ExpTime=10 seconds.

2017-11-11T00:39:01.130148 [192.168.13.109:6650] OBS>ICS go

2017-11-11T00:39:01.819764 [/dev/ttyS0] ICS>OBS STATUS: EXPSTATUS=INITIALIZING

2017-11-11T00:39:01.825648 [/dev/ttyS0] ICS>K.IC INITIALIZE 20171111.050723

2017-11-11T00:39:02.270911 [192.168.13.102:6600] K.IC>ICS DONE: INITIALIZE Initialization Complete.

2017-11-11T00:39:02.331639 [/dev/ttyS0] ICS>OBS STATUS: EXPSTATUS=ERASE
2017-11-11T00:39:02.333639 [/dev/ttyS0] ICS>K.IC ERASE (( K only ))

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2017-11-11T00:39:02.554671 [/dev/ttyS0] ICS>K.IC STATUS: AUXSTATUS ENS7=0.0 ENS6=19.2 ENS5=19.1 ENS4=41.9 ENS3=18.7 ENS2=19.1 ENS1=19.3
ENFAN=OFF ENSTAT=STANDBY CHSTAT=NC MCPOS=0 MCSTAT=STANDBY DSSTAT=NC FAPOSW=-4.222 FAPOSE=-7.187 FAPOSS=-5.631 FALIMW=0 FALIME=0
FALIMS=0 FATILTEW=-350.0 FATILTNS=-10.0 FAFOCUS=-5.680 FASTAT=STANDBY SHUTTER=CLOSED SHUTOP=STANDBY FILTER=R FILNUM=2 FILTOP=STANDBY
FSSTAT=STANDBY AUXUDATE=2017-11-11T00:39:01.654 AUXARC=Enabled AUXLINK=Up TELID=KMTS TIMESYS=UTC AUXQDATE=2017-11-11T00:39:01.706
KBUILD=KS2016-01-13:1370 MBUILD=KS2016-01-13:1370 TBUILD=KS2016-01-13:1370 NBUILD=KS2016-01-13:1370 GBUILD= ICSBUILD=KX2016-03-
23:1381 EXPSTATUS=ERASE

2017-11-11T00:39:09.575873 [192.168.13.102:6600] K.IC>ICS DONE: Erase Cycle Complete.

2017-11-11T00:39:09.595903 [/dev/ttyS0] ICS>OBS STATUS: STATUS: EXPSTATUS=INTEGRATING

2017-11-11T00:39:09.599646 [/dev/ttyS0] ICS>K.IC SHOPEN 10 OBS USESTATUS
2017-11-11T00:39:09.746911 [192.168.13.102:6600] K.IC>OBS STATUS: SHOPEN Shutter=Open (( only to sourceID ))
2017-11-11T00:39:15.399904 [192.168.13.102:6600] K.IC>OBS STATUS: SHOPEN Integration Remaining=4 sec. (( only to sourceID ))

2017-11-11T00:39:15.716673 [/dev/ttyS0] ICS>K.IC STATUS: TCSSTATUS DATE-OBS=2017-11-11T00:39:10 EXECODE=E TCS DRIVE=Disabled TCSLIMIT=No
TELMOVE=Idle AZ=0.0 ALT=90.0 SECZ=1.00 ST=05:23:41 HA=+00:00:00 EQUINOX=2000.000 DEC=-32:13:34.5 RA=05:23:07.02 TCSUDATE=2017-11-
11T00:39:01.737 TCSARC=Enabled TCSLINK=Up TIMESYS=UTC TCSQDATE=2017-11-11T00:39:02.558 EXPSTATUS=INTEGRATING

2017-11-11T00:39:15.718644 [/dev/ttyS0] ICS>K.IC FLASHNOW 20000
2017-11-11T00:39:19.997905 [192.168.13.102:6600] K.IC>ICS DONE: FLASHNOW LED Flash Done.
2017-11-11T00:39:20.064885 [192.168.13.102:6600] K.IC>OBS STATUS: FLASHNOW Shutter=Closed Integration Remaining=0 sec. (( only to
sourceID ))

2017-11-11T00:39:25.690617 [/dev/ttyS0] ICS>OBS STATUS: EXPSTATUS=READOUT
2017-11-11T00:39:25.693614 [/dev/ttyS0] ICS>K.IC GO OBS (( after GO OBS to M.IC/N.IC/T.IC ))
2017-11-11T00:39:25.747864 [192.168.13.102:6600] K.IC>ICS STATUS: GO
2017-11-11T00:39:28.404033 [192.168.13.102:6600] K.IC>OBS STATUS: GO PCTREAD=6 (( only to sourceID ))
2017-11-11T00:39:31.771026 [192.168.13.102:6600] K.IC>OBS STATUS: GO PCTREAD=17 (( only to sourceID ))
2017-11-11T00:39:35.139900 [192.168.13.102:6600] K.IC>OBS STATUS: GO PCTREAD=28 (( only to sourceID ))
2017-11-11T00:39:38.509142 [192.168.13.102:6600] K.IC>OBS STATUS: GO PCTREAD=39 (( only to sourceID ))
2017-11-11T00:39:41.878853 [192.168.13.102:6600] K.IC>OBS STATUS: GO PCTREAD=50 (( only to sourceID ))
2017-11-11T00:39:45.249016 [192.168.13.102:6600] K.IC>OBS STATUS: GO PCTREAD=61 (( only to sourceID ))
2017-11-11T00:39:48.617888 [192.168.13.102:6600] K.IC>OBS STATUS: GO PCTREAD=72 (( only to sourceID ))
2017-11-11T00:39:51.987986 [192.168.13.102:6600] K.IC>OBS STATUS: GO PCTREAD=83 (( only to sourceID ))
2017-11-11T00:39:55.356876 [192.168.13.102:6600] K.IC>OBS STATUS: GO PCTREAD=94 (( only to sourceID ))

2017-11-11T00:39:57.097874 [192.168.13.102:6600] K.IC>OBS STATUS: GO PCTREAD=100 Acquisition Complete. Disk Transfer Starting. (( only
to sourceID ))

2017-11-11T00:39:57.143816 [192.168.13.102:6600] K.IC>ICS STATUS: GO Acquisition Complete
2017-11-11T00:39:57.521652 [/dev/ttyS0] ICS>OBS DONE: EXPSTATUS=IDLE

2017-11-11T00:40:08.635972 [192.168.13.102:6600] K.IC>K.CB TRANSFER DISK0 1 ICS
2017-11-11T00:40:08.652714 [192.168.13.102:6600] K.IC>XIS PING
2017-11-11T00:40:08.652743 XIS>K.IC PONG
2017-11-11T00:40:08.697171 [192.168.13.102:6600] K.IC>ICS STATUS: GO Disk Write Complete (( after PONG response from XIS ))

2017-11-11T00:40:12.147167 [192.168.13.102:10601] K.CB>ICS DONE: Wrote LASTFILE=/mnt/ICSDData/KMTNk.20171111.050723.fits RATE=1135512
KB/sec (( form K.CB ))
2017-11-11T00:40:12.181771 [/dev/ttyS0] ICS>OBS STATUS: Wrote LASTFILE=/mnt/ICSDData/KMTNk.20171111.050723.fits RATE=1135512 KB/sec

2017-11-11T00:40:13.176526 [192.168.13.102:10601] K.CB>K.IC DONE DISK0 1
2017-11-11T00:40:13.176559 [192.168.13.102:10601] K.CB>K.IC REQ SWAP

2017-11-11T00:40:13.259921 [192.168.13.102:6600] K.IC>K.CB ACK SWAP

<< BIAS >>
2017-11-11T01:04:05.814000 [192.168.13.109:6650] OBS>ICS ledflash 1
2017-11-11T01:04:05.826645 [/dev/ttyS0] ICS>OBS DONE: LEDFLASH LEDFlashTime=1

2017-11-11T01:07:41.559850 [192.168.13.109:6650] OBS>ICS projid NEO-ENG
2017-11-11T01:07:41.574770 [/dev/ttyS0] ICS>OBS DONE: PROJID ProjID=NEO-ENG
2017-11-11T01:07:41.576759 [/dev/ttyS0] ICS>K.IC projid NEO-ENG
2017-11-11T01:07:41.652531 [192.168.13.102:6600] K.IC>ICS DONE: PROJID ProjID=NEO-ENG

2017-11-11T01:08:08.031851 [192.168.13.109:6650] OBS>ICS bias df3.bias
2017-11-11T01:08:08.046745 [/dev/ttyS0] ICS>OBS DONE: BIAS ImageType=BIAS ObjectName='df3.bias' EXP=0
2017-11-11T01:08:08.049615 [/dev/ttyS0] ICS>K.IC bias df3.bias

2017-11-11T01:08:08.159662 [192.168.13.102:6600] K.IC>ICS DONE: BIAS ImageType=BIAS ObjectName='df3.bias' EXP=0

2017-11-11T01:09:57.016941 [192.168.13.109:6650] OBS>ICS go

2017-11-11T01:09:57.722872 [/dev/ttyS0] ICS>K.IC INITIALIZE 20171111.050724

2017-11-11T01:09:58.148600 [192.168.13.102:6600] K.IC>ICS DONE: INITIALIZE Initialization Complete.

2017-11-11T01:09:58.205786 [/dev/ttyS0] ICS>OBS STATUS: EXPSTATUS=ERASE

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2017-11-11T01:09:58.206775 [/dev/ttyS0] ICS>K.IC ERASE

2017-11-11T01:09:58.457674 [/dev/ttyS0] ICS>K.IC STATUS: AUXSTATUS ENS7=0.0 ENS6=18.8 ENS5=19.1 ENS4=43.9 ENS3=18.5 ENS2=18.6 ENS1=18.6
ENFAN=OFF ENSTAT=STANDBY CHSTAT=NC MCPOS=100 MCSTAT=STANDBY DSTEL=46.0 DSALT=16.0 DSAUTO=DISABLED DSSAF=ACTIVE DSLW=CLOSED DSUP=CLOSED
DSSTAT=STANDBY FAPOSW=-4.222 FAPOSE=-7.187 FAPOSS=-5.631 FALIMW=0 FALIME=0 FALIMS=0 FATILTEW=-350.0 FATILTNS=-10.0 FAFOCUS=-5.680
FASTAT=STANDBY SHUTTER=CLOSED SHUTOP=STANDBY FILTER=R FILNUM=2 FILTOP=STANDBY FSSTAT=STANDBY AUXUDATE=2017-11-11T01:09:57.502
AUXARC=Enabled AUXLINK=Up TELID=KMTS TIMESYS=UTC AUXQDATE=2017-11-11T01:09:57.581 KBUILD=KS2016-01-13:1370 MBUILD=KS2016-01-13:1370
TBUILD=KS2016-01-13:1370 NBUILD=KS2016-01-13:1370 GBUILD= ICSBUILD=KX2016-03-23:1381 EXPSTATUS=ERASE

2017-11-11T01:10:05.447635 [192.168.13.102:6600] K.IC>ICS DONE: Erase Cycle Complete.

2017-11-11T01:10:05.584668 [/dev/ttyS0] ICS>K.IC STATUS: TCSSTATUS DATE-OBS=2017-11-11T01:10:06 EXECODE=e TCSDRIVE=Enabled TCSLIMIT=No
TELMOVE=Idle AZ=81.0 ALT=46.0 SECZ=1.39 ST=05:54:42 HA=-03:24:47 EQUINOX=2000.000 DEC=-28:18:15.9 RA=09:18:54.68 TCSUDATE=2017-11-
11T01:09:57.887 TCSARC=Enabled TCSLINK=Up TIMESYS=UTC TCSQDATE=2017-11-11T01:09:58.457 EXPSTATUS=INTEGRATING

2017-11-11T01:10:11.484827 [/dev/ttyS0] ICS>OBS STATUS: Shutter=Closed Integration Remaining=0 sec. EXPSTATUS=INTEGRATING

2017-11-11T01:10:11.577770 [/dev/ttyS0] ICS>OBS STATUS: EXPSTATUS=READOUT

2017-11-11T01:10:11.581769 [/dev/ttyS0] ICS>K.IC GO OBS

2017-11-11T01:10:11.632585 [192.168.13.102:6600] K.IC>ICS STATUS: GO

2017-11-11T01:10:14.288744 [192.168.13.102:6600] K.IC>OBS STATUS: GO PCTREAD=6

2017-11-11T01:10:17.632648 [192.168.13.102:6600] K.IC>OBS STATUS: GO PCTREAD=17

2017-11-11T01:10:21.024775 [192.168.13.102:6600] K.IC>OBS STATUS: GO PCTREAD=28

2017-11-11T01:10:24.393594 [192.168.13.102:6600] K.IC>OBS STATUS: GO PCTREAD=39

2017-11-11T01:10:27.763742 [192.168.13.102:6600] K.IC>OBS STATUS: GO PCTREAD=50

2017-11-11T01:10:31.132665 [192.168.13.102:6600] K.IC>OBS STATUS: GO PCTREAD=61

2017-11-11T01:10:34.502657 [192.168.13.102:6600] K.IC>OBS STATUS: GO PCTREAD=72

2017-11-11T01:10:37.871786 [192.168.13.102:6600] K.IC>OBS STATUS: GO PCTREAD=83

2017-11-11T01:10:41.240631 [192.168.13.102:6600] K.IC>OBS STATUS: GO PCTREAD=94

2017-11-11T01:10:42.981594 [192.168.13.102:6600] K.IC>OBS STATUS: GO PCTREAD=100 Acquisition Complete. Disk Transfer Starting.

2017-11-11T01:10:43.028680 [192.168.13.102:6600] K.IC>ICS STATUS: GO Acquisition Complete

2017-11-11T01:10:43.455637 [/dev/ttyS0] ICS>OBS DONE: EXPSTATUS=IDLE

2017-11-11T01:10:52.349919 [192.168.13.102:6600] K.IC>K.CB TRANSFER DISK1 1 ICS

2017-11-11T01:10:52.366681 [192.168.13.102:6600] K.IC>XIS PING

2017-11-11T01:10:52.366716 XIS>K.IC PONG

2017-11-11T01:10:52.410640 [192.168.13.102:6600] K.IC>ICS STATUS: GO Disk Write Complete

2017-11-11T01:10:54.466056 [192.168.13.102:10601] K.CB>ICS DONE: Wrote LASTFILE=/mnt/ICSDData/KMTNk.20171111.050724.fits RATE=1165665
KB/sec

2017-11-11T01:10:54.507636 [/dev/ttyS0] ICS>OBS STATUS: Wrote LASTFILE=/mnt/ICSDData/KMTNk.20171111.050724.fits RATE=1165665 KB/sec

2017-11-11T01:10:55.352128 [192.168.13.102:10601] K.CB>K.IC DONE DISK1 1

2017-11-11T01:10:55.352284 [192.168.13.102:10601] K.CB>K.IC REQ SWAP

2017-11-11T01:10:55.437021 [192.168.13.102:6600] K.IC>K.CB ACK SWAP

문서 업데이트 내용 및 시험결과 정리

20190710 업데이트 - 내용정리 + 문장수정

20190624 K.IC 개별구동 명령어 <Example> 추가

20180315 command test at SAAO

- ICS에서 파일번호 설정 명령어 EXPNUM 잘 동작함.

- K.IC에서 EXPNUM --> 4-digit fixed,

- K.IC에서 파일명 지정은 INITIALIZE xxxxx.xxxx 사용해야 함.

INITIALIZE 명령어로 날짜.번호 전체 설정 가능. (KMTNk.(여기설정).fits)

매개변수를 넣지 않으면 yyyyymmdd.nnnn으로 설정됨.

nnnn은 expnum로 변경가능 하지만 4-digit으로 고정.

- LEDFLASH 시험결과, 어떻게 해도 영상 잘나옴.

OSU에서 안나온 이유 검토/디버깅 필요 (IC2.img 파일 버전문제가 의심됨)

- LEDFLASH 0로 해도 문제 없는 듯

--> BIAS 영상만 확인, sky 영상도 확인해봐야 함.

--> 최근 사이트에서 LEDFLASH 0으로 해오고 있는데 문제 없는 듯.

- OBSAgent에 ICS commands 그룹으로 ACQSTATUS, EXPNUM 추가해야 함

EOF